KO 1 3.4.1 | Operational Objectives

AQA A-LEVEL BUSINESS 7132 - UNIT 3.4 OPERATIONAL MANAGEMENT

WHAT AQA LISTS

Costs - e.g. reduce unit costs by 5%.

Quality - e.g. cut the number of faulty products or complaints.

Speed of response - how quickly the business can meet orders or react to changes.

Flexibility - the ability to change the volume or range of products.

Environmental objectives - e.g. cut waste, energy use or emissions.

Added value - increase the difference between the price and the cost of bought-in goods and services.

ADDED VALUE

Added value = sales revenue - cost of bought-in goods and services

It can be increased by raising the price customers will pay (better design, branding, service) or by reducing the cost of bought-in materials and services.

WHY SET OPERATIONAL OBJECTIVES?

They give the operations team clear targets and a way of measuring performance.

They help coordinate operations with marketing, finance and HR.

They should support the corporate objectives, e.g. a low-cost strategy needs cost objectives; a premium brand needs quality objectives.

INFLUENCES AND TRADE-OFFS

Influences: corporate objectives, the market and competition, technology, the resources available, and ethical and environmental pressures.

Objectives can conflict. Cutting costs may reduce quality; faster response may need more inventory, which raises costs.

EXAM TRAP

Added value is not the same as profit. It takes away only the cost of bought-in goods and services, not other costs such as wages and rent.

ANALYSIS CHAIN

Objective to cut energy use → investment in efficient machinery → lower running costs and emissions → lower unit costs and a better environmental image.

TOP GRADE CONNECTION

Which objective matters most depends on how the business competes. A budget airline puts cost first; a bespoke furniture maker puts quality and flexibility first.

KO 2 3.4.2 | Operations Data

AQA A-LEVEL BUSINESS 7132 - UNIT 3.4 OPERATIONAL MANAGEMENT

CALCULATIONS

Labour productivity = output over a time period ÷ number of employees

Unit cost (average cost) = total costs ÷ number of units of output

Capacity = the maximum output a business can produce in a given period with its current resources.

Capacity utilisation (%) = actual output ÷ maximum possible output × 100

WORKED EXAMPLE

A factory can make 2,000 units a week. It makes 1,500 units with 25 staff. Total costs are £30,000 a week.

Capacity utilisation = 1,500 ÷ 2,000 × 100 = 75%

Labour productivity = 1,500 ÷ 25 = 60 units per employee per week

Unit cost = £30,000 ÷ 1,500 = £20

USING THE DATA

Compare figures over time and with competitors to spot strengths and weaknesses.

Higher labour productivity usually lowers labour cost per unit.

Higher capacity utilisation spreads fixed costs over more units, lowering unit costs.

Operations data helps decide whether to invest, change staffing or change production methods.

INTERPRETING WITH CARE

Productivity figures ignore quality - faster output with more faults is not an improvement.

Unit costs depend on output. If output falls, fixed costs are spread over fewer units and unit cost rises.

One figure in isolation says little; look at trends and the reasons behind them.

EXAM TRAP

Always give units and £ signs: 60 units per employee per week, £20 per unit, 75%. A bare number loses marks.

ANALYSIS CHAIN

Capacity utilisation rises from 60% to 85% → fixed costs spread over more units → unit cost falls → business can cut price or earn a higher margin.

TOP GRADE CONNECTION

Ask why a figure has changed. Productivity can rise because of better training, or because staff are being pushed harder in a way that will raise absence and turnover later.

KO 3 3.4.3 | Capacity

AQA A-LEVEL BUSINESS 7132 - UNIT 3.4 OPERATIONAL MANAGEMENT

WHY CAPACITY MATTERS

Capacity sets the maximum a business can supply. Too little means lost sales; too much means paying for resources that are not used.

Capacity utilisation affects unit costs because fixed costs are spread over the units produced.

UNDER-UTILISATION

Low utilisation means high fixed costs per unit and possibly idle staff with low morale.

Ways to improve: increase sales (promotion, new markets), make products for other firms, rent out spare space, or reduce capacity (sell assets, close sites, cut shifts).

VERY HIGH UTILISATION

Running close to 100% gives low unit costs, but leaves no time for maintenance, little room for extra orders and puts pressure on staff.

Quality can suffer and orders may be turned away.

Options: extra shifts, subcontracting work, temporary staff, or investing in more capacity.

USING CAPACITY EFFICIENTLY

Match capacity to expected demand, including seasonal patterns.

Use flexible staff and outsourcing to cope with peaks rather than carrying permanent spare capacity.

Technology and better scheduling can increase capacity without major investment.

EXAM TRAP

100% is not the ideal capacity utilisation. Very high utilisation brings its own problems - no slack for maintenance, breakdowns or rush orders.

ANALYSIS CHAIN

Utilisation falls as demand drops → fixed costs spread over fewer units → unit cost rises → margins squeezed → pressure to cut capacity or win new orders.

TOP GRADE CONNECTION

Whether low utilisation is a problem depends on the cause. A temporary seasonal dip may be best handled with flexible staff, while a long-term fall in demand may justify closing capacity.

KO 4 3.4.3 | Efficiency, Productivity & Technology

AQA A-LEVEL BUSINESS 7132 - UNIT 3.4 OPERATIONAL MANAGEMENT

EFFICIENCY AND PRODUCTIVITY

Efficiency - producing output with the fewest resources, i.e. at the lowest unit cost.

Labour productivity - output per worker in a period.

Higher productivity lowers unit labour costs, which can mean lower prices or higher margins.

INCREASING EFFICIENCY AND PRODUCTIVITY

Training staff and improving motivation.

Better equipment and technology.

Improving the way work is organised, e.g. layout, scheduling, removing wasted steps.

Lean production methods.

DIFFICULTIES

Investment in equipment and training costs money and takes time to pay back.

Staff may resist new methods or fear job losses.

Pushing for speed can damage quality and morale.

LABOUR OR CAPITAL INTENSIVE?

Labour intensive - relies mainly on people, e.g. a hairdresser or handmade goods. Flexible and suits personal service, but labour costs are high.

Capital intensive - relies mainly on machinery and technology, e.g. a car factory. Low unit costs at high volume, but high fixed costs and less flexible.

The optimal mix depends on the product, volume, costs of labour and capital, and what customers value. Automation and robotics are pushing many firms towards capital.

EXAM TRAP

Productivity and production are not the same. Production is total output; productivity is output per worker. Production can rise while productivity falls.

ANALYSIS CHAIN

Investment in automation → more output per worker → lower labour cost per unit → lower prices or higher margins → more competitive, once the cost of the machinery is recovered.

TOP GRADE CONNECTION

Technology is not always the answer. For a business competing on personal service or bespoke products, a labour-intensive approach may be what customers are paying for.

KO 5 3.4.3 | Lean Production & Just in Time

AQA A-LEVEL BUSINESS 7132 - UNIT 3.4 OPERATIONAL MANAGEMENT

LEAN PRODUCTION

An approach that aims to cut waste of all kinds - materials, time, space, effort and inventory - while maintaining quality.

Staff are often involved in finding improvements.

JUST IN TIME (JIT)

Materials arrive just as they are needed and products are made to order, so little inventory is held.

Benefits - less cash tied up in inventory, less storage space, less waste and damage, problems are spotted quickly.

Difficulties - relies on very reliable suppliers, no buffer if demand rises suddenly, frequent deliveries cost more, any disruption can stop production.

JUST IN CASE

Holding buffer inventory in case of rising demand or supply problems.

Benefits - can meet sudden orders, protected against supply disruption, may get bulk-buying discounts.

Drawbacks - cash tied up, storage costs, risk of waste, damage or items going out of date.

WHICH TO CHOOSE?

JIT suits stable demand, reliable local suppliers and products that are costly to store.

Just in case suits unpredictable demand, long or unreliable supply chains, and situations where running out would be very costly.

Many businesses use a mixture for different materials.

EXAM TRAP

JIT does not mean holding no inventory at all and never running out. It reduces inventory, which increases the risk of running out if supply is disrupted.

ANALYSIS CHAIN

JIT introduced → less inventory held → less cash tied up and lower storage costs → better cash flow, but a supplier delay can stop production.

TOP GRADE CONNECTION

Supply chain shocks have shown the risk of very lean systems. Judge JIT by how reliable the supply chain is and how costly a stoppage would be for the business.

KO 6 3.4.4 | Quality

AQA A-LEVEL BUSINESS 7132 - UNIT 3.4 OPERATIONAL MANAGEMENT

QUALITY CONTROL AND QUALITY ASSURANCE

Quality control - checking products at the end (or at stages) of production to find and remove faults. Catches faults but does not stop them happening, and waste is created.

Quality assurance - building quality into every stage so that faults are prevented, with staff responsible for their own work. Less waste, but needs training and a culture where staff take responsibility.

METHODS OF IMPROVING QUALITY

Training staff and giving them responsibility for quality.

Better materials and more reliable suppliers.

Better equipment and technology.

Setting clear standards and measuring performance against them.

BENEFITS AND DIFFICULTIES

Benefits - fewer complaints and returns, less waste, stronger reputation, customer loyalty, ability to charge higher prices.

Difficulties - training and equipment cost money, results take time, staff may resist extra checks or responsibility.

CONSEQUENCES OF POOR QUALITY

Waste and the cost of reworking or scrapping products.

Returns, refunds and possible legal claims.

Damage to reputation, lost customers and lower sales.

In serious cases, product recalls and fines.

EXAM TRAP

Quality does not mean luxury. It means meeting customers' expectations consistently - a budget product can be high quality for its price.

ANALYSIS CHAIN

Move from quality control to quality assurance → faults prevented rather than found → less waste and rework → lower unit costs and fewer complaints → stronger reputation.

TOP GRADE CONNECTION

Improving quality costs money now and pays back later. Judge whether customers in this market will reward higher quality with loyalty or higher prices.

KO 7 3.4.5 | Matching Supply to Demand

AQA A-LEVEL BUSINESS 7132 - UNIT 3.4 OPERATIONAL MANAGEMENT

WHY IT MATTERS

If supply is below demand, sales are lost and customers may go to rivals.

If supply is above demand, the business is left with unsold inventory or spare capacity and higher unit costs.

Demand is often uneven - seasonal, daily or unpredictable - so firms need ways to adjust.

OUTSOURCING

Using another business to carry out an activity or make part of the output.

Lets the business meet peaks in demand without investing in extra capacity.

Less control over quality and delivery, and the supplier takes a profit margin.

TEMPORARY AND PART-TIME STAFF

Taken on to cover busy periods or particular hours, e.g. Christmas retail staff.

Keeps labour costs flexible, but temporary staff may be less trained or committed, which can affect service and quality.

PRODUCING TO ORDER

Products are made only once an order is received.

No unsold inventory and products can be tailored to customers, but customers must wait and output can be uneven.

Other options: building up inventory in quiet periods, and using price and promotion to shift demand to quieter times.

EXAM TRAP

Matching supply to demand is about flexibility, not simply producing more. Over-supply is as costly as under-supply.

ANALYSIS CHAIN

Seasonal peak covered with temporary staff → output matches demand without permanent extra costs → sales captured and unit costs controlled → but service may slip if temps are poorly trained.

TOP GRADE CONNECTION

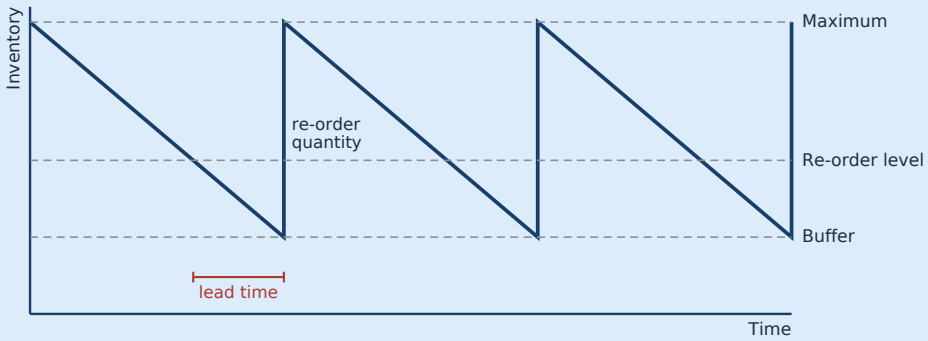
The best method depends on the product and the pattern of demand. Producing to order works for bespoke furniture, not for fresh bread on a Saturday morning.

KO 8 3.4.5 | Inventory Control

AQA A-LEVEL BUSINESS 7132 - UNIT 3.4 OPERATIONAL MANAGEMENT

INVENTORY CONTROL CHART

Shows inventory falling as it is used and rising when a delivery arrives.



KEY TERMS

Lead time - the time between placing an order and the delivery arriving.

Re-order level - the inventory level at which a new order is placed.

Buffer inventory - the minimum level kept in case of delays or extra demand.

Re-order quantity - the amount ordered each time.

Maximum level - buffer inventory + re-order quantity.

READING THE CHART

The steeper the line falls, the faster inventory is being used.

Re-order level = buffer inventory + (usage per period × lead time).

If a delivery is late or usage rises, inventory falls into the buffer.

INFLUENCES ON THE AMOUNT HELD

Reliability and lead time of suppliers.

How predictable demand is.

Cost of storage and of cash tied up.

Whether the item is perishable or quickly goes out of date.

Bulk-buying discounts and the cost of running out.

EXAM TRAP

Buffer inventory is not the re-order level. The re-order level is above the buffer so that the delivery arrives before the buffer is used.

ANALYSIS CHAIN

Supplier lead time increases → re-order level must rise → more inventory held on average → more cash tied up and higher storage costs.

TOP GRADE CONNECTION

Holding less inventory improves cash flow but increases the risk of running out. Judge the right level from the cost of a stock-out compared with the cost of holding inventory.

KO 9 3.4.5 | Suppliers & Supply Chains

AQA A-LEVEL BUSINESS 7132 - UNIT 3.4 OPERATIONAL MANAGEMENT

CHOOSING SUPPLIERS

Price and payment terms.

Quality and reliability of delivery.

Capacity and flexibility to meet changes in orders.

Location and lead time.

Ethical and environmental standards.

MANAGING THE SUPPLY CHAIN

The supply chain is every stage from raw materials to the final customer.

Managing it well means good communication with suppliers, sharing information on demand, monitoring performance and building long-term relationships.

Technology helps: online ordering, real-time inventory tracking and data shared with suppliers.

VALUE OF GOOD SUPPLY CHAIN MANAGEMENT

Lower costs and less inventory.

Faster, more reliable delivery to customers.

Better quality and fewer disruptions.

Can be a source of competitive advantage that rivals find hard to copy.

THE VALUE OF OUTSOURCING

For - access to specialist skills, lower costs, more flexible capacity, the business can focus on what it does best.

Against - loss of control over quality and timing, dependence on another business, possible damage to reputation if the supplier behaves unethically.

EXAM TRAP

The cheapest supplier is not automatically the best. Unreliable delivery or poor quality can cost far more than the saving on price.

ANALYSIS CHAIN

Long-term partnership with a key supplier → shared demand forecasts → supplier plans capacity in advance → fewer delays → better delivery to customers and less buffer inventory needed.

TOP GRADE CONNECTION

Ethical and environmental standards in the supply chain increasingly affect reputation. A low-cost supplier with poor working conditions can become a serious risk for the brand.